

624 Trees: Learning from the Central Park Squirrel Census

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0.1 Why Use Tree Methods on Squirrel Data?

Tree models are helpful here because the squirrel census is full of mixed data types: categorical fur colors, yes/no behaviors, vague human interaction signals, and spatial labels such as hectares. A single tree can turn that messy field data into readable splits, while ensembles let us check whether the same patterns hold up when we average across many trees.

Instead of treating this as a generic modeling exercise, I use tree methods as a field guide:

1. A classification tree asks where squirrels are likely to be observed: on the ground or above it.
2. A regression tree asks how high squirrels perch once they are already above ground.
3. A rule-based tree turns feeding behavior into short, interpretable rules.
4. A bagged tree checks whether an ensemble improves the stability of the location prediction.

0.2 Data Snapshot

```
kable(overview, caption = "Quick overview of the squirrel census data used in this  
↪ report.")
```

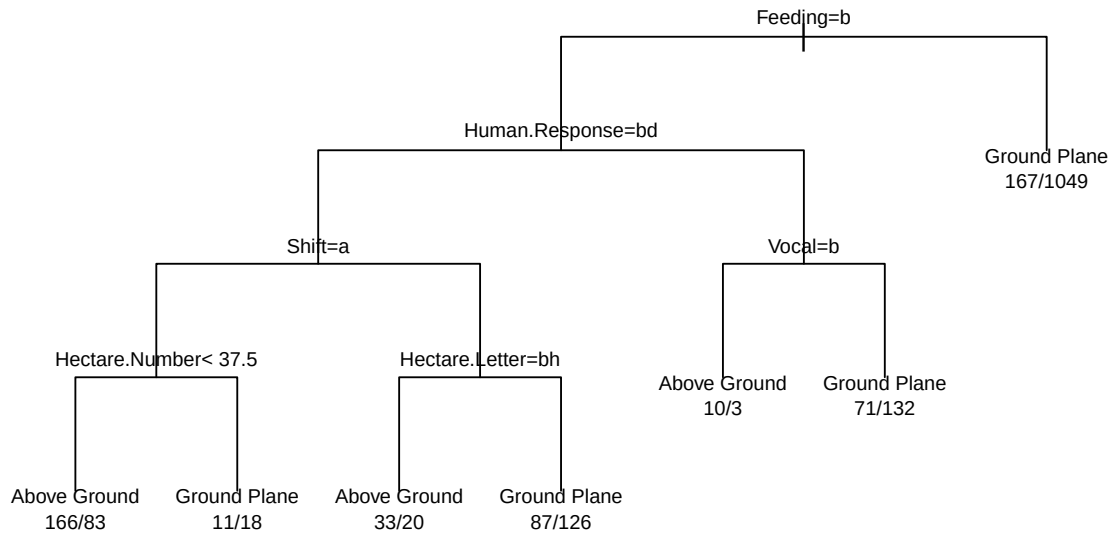
Table 1: Quick overview of the squirrel census data used in this report.

Measure	Value
Total observations	3023
Distinct hectares	339
Ground-plane sightings	2116
Above-ground sightings	843
Sightings with recorded height	793

0.3 Classification Tree: What Predicts Ground vs. Above-Ground Sightings?

This tree treats the park like a behavioral landscape. The goal is to see whether movement, feeding, vocalization, tail signals, and basic location cues help separate squirrels seen on the ground from squirrels spotted above ground.

```
location_tree <- rpart(Location.Class ~ ., data = train_location, method = "class")
plot(location_tree, uniform = TRUE, margin = 0.08)
text(location_tree, use.n = TRUE, cex = 0.7)
```



```
location_pred <- predict(location_tree, test_location, type = "class")
location_metrics <- classification_metrics(test_location$Location.Class,
  ↪ location_pred)
location_importance <- top_importance(location_tree)
```

```
kable(location_metrics, caption = "Classification tree performance for predicting
↪ squirrel location.")
```

Table 2: Classification tree performance for predicting squirrel location.

Accuracy	Kappa
0.749	0.271

```
kable(location_importance, caption = "Most influential variables in the location
↪ tree.")
```

Table 3: Most influential variables in the location tree.

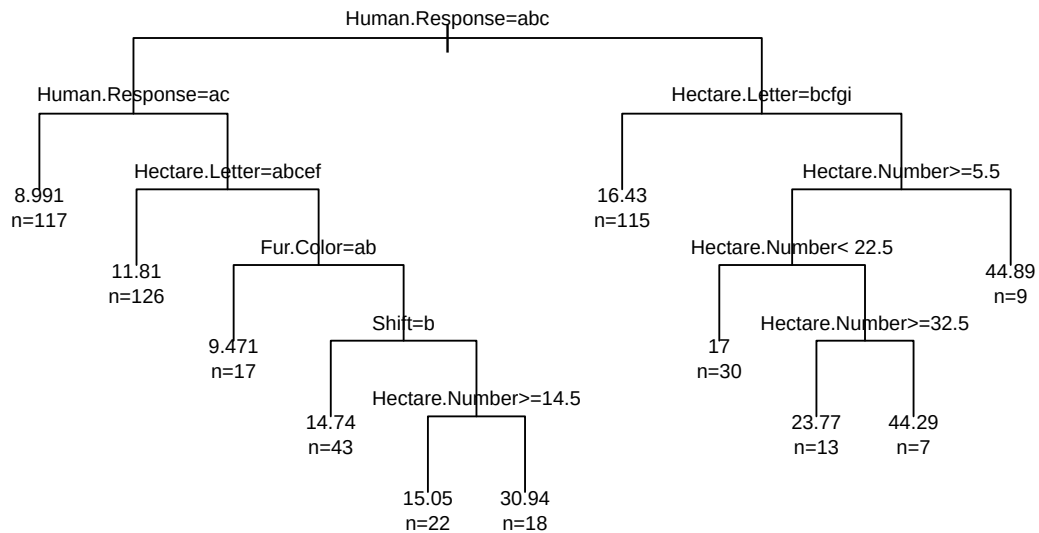
Variable	Importance
Feeding	121.25
Active	56.93
Human.Response	12.43
Shift	9.36
Vocal	6.21
Hectare.Number	5.25

The classification tree helps answer a practical question for field observation: what makes a squirrel more likely to be seen above ground? In this data, active movement and feeding-related behavior matter much more than fur color, which suggests behavior is more informative than appearance for understanding where squirrels position themselves.

0.4 Regression Tree: How High Do Above-Ground Squirrels Perch?

Once a squirrel is above ground, the next question is not just *whether* it climbed, but *how high*. This regression tree uses the same mix of spatial and behavioral cues to estimate recorded height.

```
height_tree <- rpart(Above.Ground.Height ~ ., data = train_height, method = "anova")
plot(height_tree, uniform = TRUE, margin = 0.08)
text(height_tree, use.n = TRUE, cex = 0.7)
```



```
height_pred <- predict(height_tree, test_height)
height_metrics <- regression_metrics(test_height$Above.Ground.Height, height_pred)
height_importance <- top_importance(height_tree)

kable(height_metrics, caption = "Regression tree performance for above-ground
↪ height.")
```

Table 4: Regression tree performance for above-ground height.

RMSE	MAE
12.42	9.11

```
kable(height_importance, caption = "Most influential variables in the height tree.")
```

Table 5: Most influential variables in the height tree.

Variable	Importance
Hectare.Number	10877.62
Human.Response	8211.92
Hectare.Letter	6634.14
Shift	1497.91
Fur.Color	1370.95

Variable	Importance
Tail.Signal	331.20

This model treats height as a kind of vertical habitat choice. The most important predictors are park position and human-response context rather than simple color categories, which hints that where a squirrel is seen in the park may be more useful than how it looks when we want to explain height.

0.5 Rule-Based Tree: What Simple Rules Describe Feeding Behavior?

Rule models are useful when we want a short, readable description instead of a large branching diagram. Here, I predict whether a squirrel is feeding or not feeding from its activity level, location, and other observation cues.

```
feeding_rules <- C5.0(Feeding ~ ., data = train_feeding, rules = TRUE)
feeding_pred <- predict(feeding_rules, test_feeding)
feeding_metrics <- classification_metrics(test_feeding$Feeding, feeding_pred)
feeding_usage <- as.data.frame(C5imp(feeding_rules, metric = "usage")) %>%
  rownames_to_column(var = "Variable") %>%
  rename(Usage = Overall) %>%
  arrange(desc(Usage)) %>%
  mutate(Usage = round(Usage, 2))

kable(feeding_metrics, caption = "Rule-based tree performance for feeding behavior.")
```

Table 6: Rule-based tree performance for feeding behavior.

Accuracy	Kappa
0.778	0.55

```
kable(head(feeding_usage, 6), caption = "Most-used predictors in the feeding rules.")
```

Table 7: Most-used predictors in the feeding rules.

Variable	Usage
Active	100
Shift	0
Age	0
Fur.Color	0
Hectare.Number	0
Hectare.Letter	0

```
summary(feeding_rules)
```

```
##
## Call:
## C5.0.formula(formula = Feeding ~ ., data = train_feeding, rules = TRUE)
```

```

##
##
## C5.0 [Release 2.07 GPL Edition]      Wed Apr  1 21:30:42 2026
## -----
##
## Class specified by attribute `outcome'
##
## Read 1976 cases (11 attributes) from undefined.data
##
## Rules:
##
## Rule 1: (1019/102, lift 1.5)
##   Active = Still
##   ->  class Feeding   [0.899]
##
## Rule 2: (957/298, lift 1.8)
##   Active = Active
##   ->  class Not feeding [0.688]
##
## Default class: Feeding
##
##
## Evaluation on training data (1976 cases):
##
##           Rules
##   -----
##      No      Errors
##
##      2  400(20.2%)  <<
##
##      (a)  (b)      <-classified as
##   ----  ----
##      917  298      (a): class Feeding
##      102  659      (b): class Not feeding
##
##
## Attribute usage:
##
## 100.00% Active
##
## Time: 0.0 secs

```

The rules are surprisingly compact. In this dataset, activity level does a lot of the work: squirrels that are still are much more often recorded as feeding, while squirrels that are actively running, climbing, or chasing are more often recorded as not feeding. That is exactly the kind of ecological shorthand tree methods are good at surfacing.

0.6 Bagged Trees: Does an Ensemble Improve the Location Story?

A single tree is easy to interpret, but it can be sensitive to how the sample is split. Bagging averages over many trees, which usually gives a more stable answer. I apply bagging to the same ground-vs-above-ground problem

used in the first section.

```
bag_model <- bagging(Location.Class ~ ., data = train_location, nbagg = 25)
bag_pred <- predict(bag_model, test_location)

if (is.list(bag_pred) && "class" %in% names(bag_pred)) {
  bag_pred <- bag_pred$class
}

bag_pred <- factor(bag_pred, levels = levels(test_location$Location.Class))

bag_metrics <- classification_metrics(test_location$Location.Class, bag_pred) %>%
  mutate(Model = "Bagged trees")

single_tree_metrics <- location_metrics %>%
  mutate(Model = "Single tree")

comparison <- bind_rows(single_tree_metrics, bag_metrics) %>%
  select(Model, Accuracy, Kappa)

kable(comparison, caption = "Comparing a single location tree with a bagged ensemble.")
```

Table 8: Comparing a single location tree with a bagged ensemble.

Model	Accuracy	Kappa
Single tree	0.749	0.271
Bagged trees	0.739	0.319

The bagged model is only slightly more accurate here, but that comparison is still useful. It suggests that the strongest patterns in this squirrel census are already fairly stable, so the simple tree captures much of the signal while the ensemble adds a small amount of extra reliability.

0.7 Conclusion

Tree methods help turn this squirrel census into more than a table of observations. They show that:

1. Behavior is more informative than fur color for predicting where squirrels are seen.
2. Above-ground height looks partly like a spatial decision, not just a random measurement.
3. Feeding behavior can be summarized with a small set of readable rules.
4. Bagging confirms that the main location patterns are real enough to survive beyond a single tree.

That is the main value of tree-based models in a dataset like this one: they do not just predict outcomes, they organize messy field observations into a form that is easier to explain.